浙江大学爱丁堡大学联合学院

ZJU-UoE Institute

Introduction to Programming 2

IBI 1, Lecture 4.2

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Semester 2, 2025/26

Now…

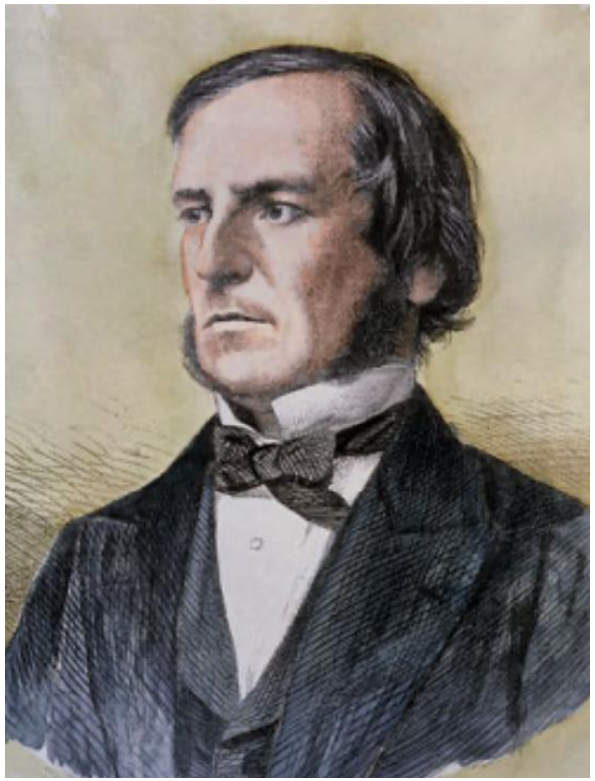
- Define re-usable functions.
- Count and keep track of numbers.
- Compare things to other things.
- Include decision points.
- Repeat things until something happens.

Outline

1. Booleans
2. If statements
3. Loops
4. Paired programming

Introduction to Booleans

- Boolean variables (= Booleans) are special variables.
- They can take only two values, "True" and "False".
- Booleans are useful when something needs to be checked to determine how the programme continues, e.g. check whether there is a turn in the track or whether it goes straight, adjust running direction accordingly.



George Boole

Comparing values

- Booleans are often used for comparison.
- Is a smaller than b?

```
a < b
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  `a >= b`
- Is a equal to b?

  `a == b`

  Why two equal signs?

Comparing values

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  `a == b`

  One equal sign would assign the value of variable b to a
- Is a not equal to b?

  `a != b`

Truth tables

Booleans can be combined and modified using not, and, and or. If A is True, then not A is False, and the other way round. The meanings of and and or can be seen from these truth tables.

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True	False	False
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True	False	False
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Or		
A	B	A or B
True	True	True
True	False	True
False	True	True
False	False	False

Exercises

Determine the truth of the following statements

- `not (4>0)`
- `1 <= 1`
- `(5 < 7) and (12 < 4)`
- `3*4 == 12 or 2 < 5`
- `x = 5`

```
y = 5
x == y
```

Different kinds of "or"

Sometimes, we need "either x or y" (i.e. x or y, but not both). What would the truth table for this look like? Can you write a line of Python code to express this?

Exercises

Determine the truth of the following statements

- `not (4>0)` → `False`
- `1 <= 1` → `True`
- `(5 < 7) and (12 < 4)` → `False`
- `3*4 == 12 or 2 < 5` → `True`
- `x = 5`

`y = 5`

`x == y` → `True`

Different kinds of "or"

Sometimes, we need "either x or y" (i.e. x or y, but not both). What would the truth table for this look like? Can you write a line of Python code to express this? `(x == True or y == True) and x != y`

Outline

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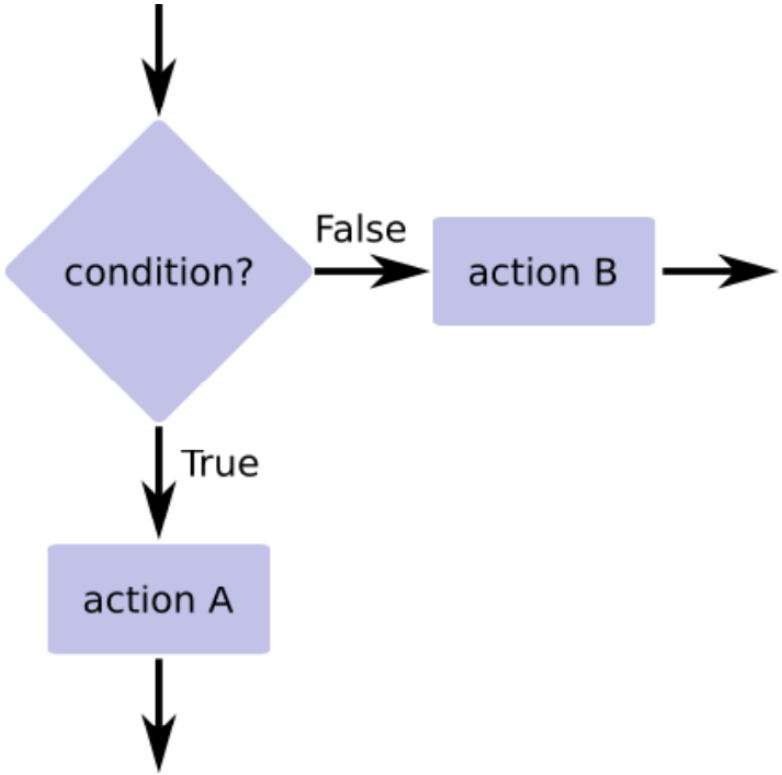
If statements

If-statements are used when a decision has to be made before the programme goes on.



If statements

If

```
if x%2 == 0:
      print ("x is even")
```

- `print()` – show something on the screen.
- All the lines that are indented belong to the same `if` statement (in this case, just one line).

If statements

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What would this return for x=4, x=3 and x=3.14?

If statements

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What would this return for x=4, x=3 and x=3.14?

True, False, False

If statements

If – Else

```
if x%2 == 0:
      print ("x is even")
else:
      print("x is odd")
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If statements

If – Else

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if x%2 == 0:
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If – Elif (Else if) – Else

```
if x%2 == 0:
     print ("x is even")
elif x%2 == 1:
     print("x is odd")
else:
     print("x is not an integer")
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If statements

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If statements

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if x%2 == 0:
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```

What would this return for x=4, x=3 and x=3.14?

Even, odd, not an integer

If Statements

Collatz sequence

For any integer c_n, the next number n the Collatz sequence c_{n+1} is defined as:

$$c_{n+1} \begin{cases} \frac{c_n}{2} & \text{if cn is even} \\ 3 * c_n + 1 & \text{if cn is odd} \end{cases}$$

Can you write a piece of code that takes any integer c_n and computes the next number in the Collatz sequence?

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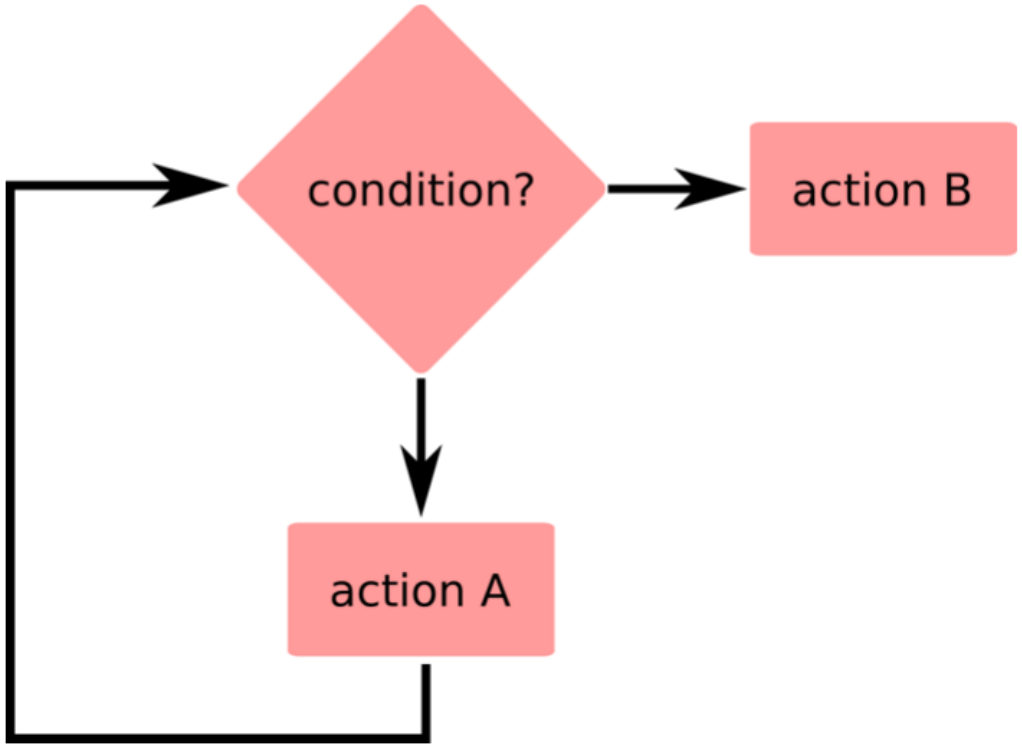
Loops

- Loops allow users to repeat the same thing many times.
- Repetition can happen either a pre-determined number of times or until a pre-determined condition is fulfilled.

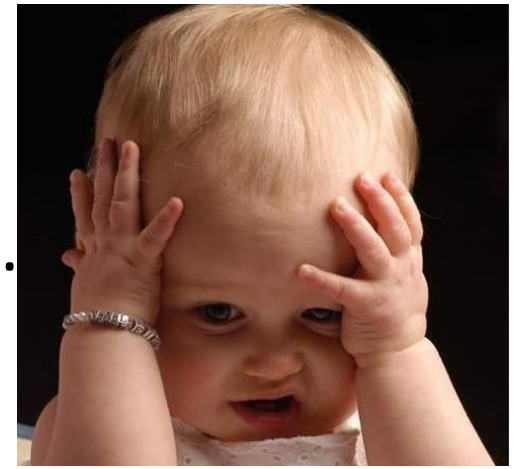


For-loop

- Do something a specific number of times.
- e.g. count down from 25:

```
count=25
print(count)
for i in range(1,25):
    count=count-1
    print("count: ", count)
```

- `range(a,b)` – numbers from a (included) to b (excluded).
- All the lines that are indented belong to the same for statement (two in this case).



For-loop

Excercise

What is the output of this piece of code?

```
for j in range(0,10):
     print("2*", j, " is ", 2*j)

print(":)")
```

For-loop

Excercise

What is the output of this piece of code?

```
for j in range(0,10):
     print("2*", j, " is ", 2*j)

print(":)")
```

2*0 is 0, 2*1 is 1,…, 2*9 is 18

:)

While-loop

- Do something until something happens.

While-loop

- Do something until something happens.
- e.g. draw random numbers until a specific number is drawn.

```
import random
j=0
while j!= 23:
      j = random.randint(1,100)
      print(j)
```

- random – library for random numbers
- randint(a,b) – draw random integer between a and b

While-loop

- Alternative:

```
import random
while 1==1:
	j = random.randint(1,100)
	print(j)
	if j==23:
		break
```

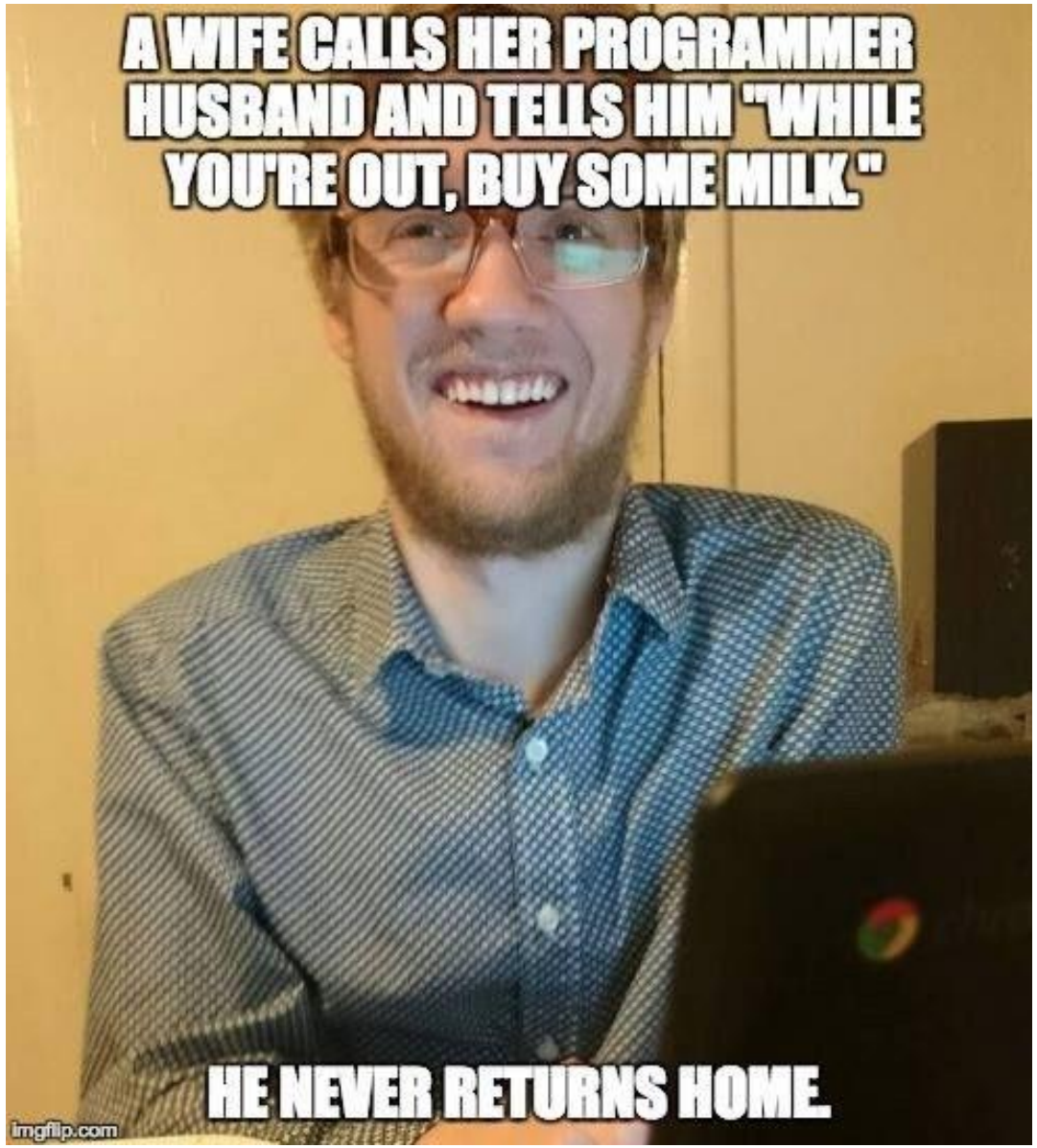
While-loop

- Alternative:

```
import random
while 1==1:
	j = random.randint(1,100)
	print(j)
	if j==23:
	break
```

- Why are "while" loops dangerous?

While-loop


A WIFE CALLS HER PROGRAMMER HUSBAND AND TELLS HIM "WHILE YOU'RE OUT, BUY SOME MILK."
HE NEVER RETURNS HOME
imgflip.com

Outline

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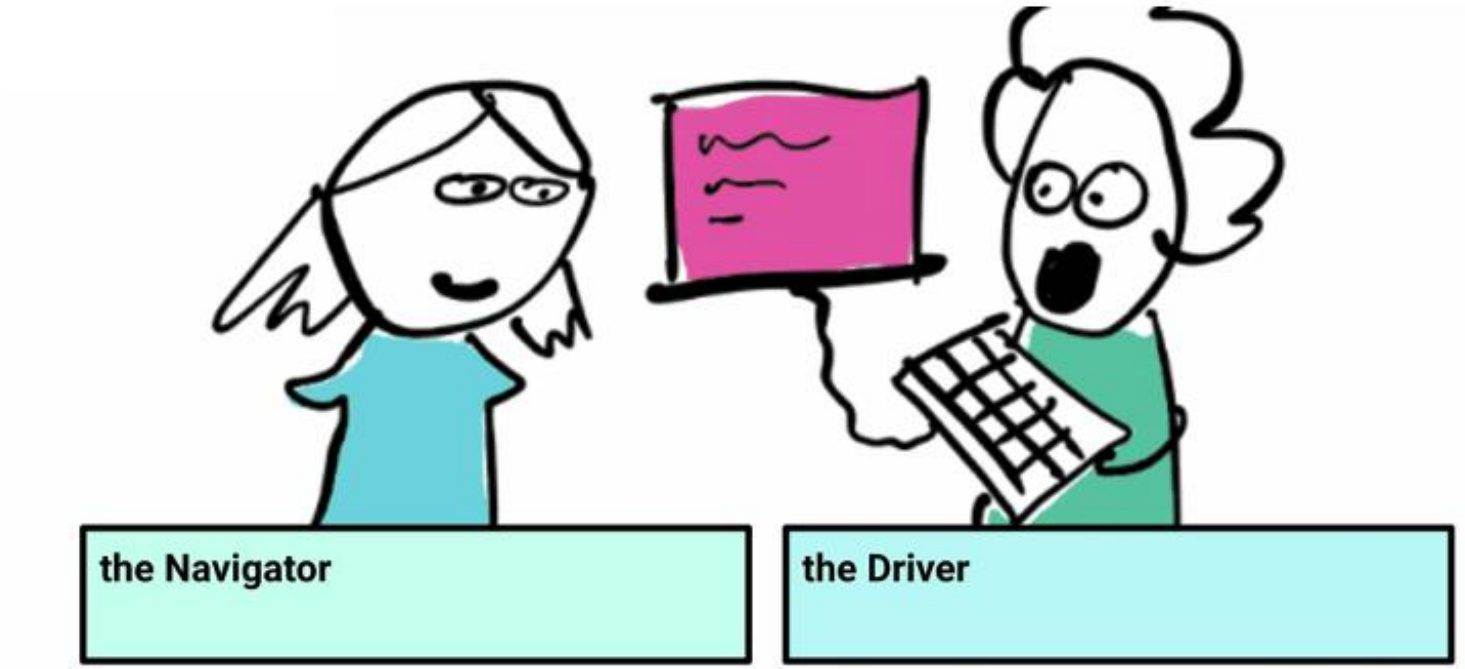
Paired programming

- The driver: types at the keyboard, explaining what they are doing
- The navigator: helps out, looking for bugs, asking questions

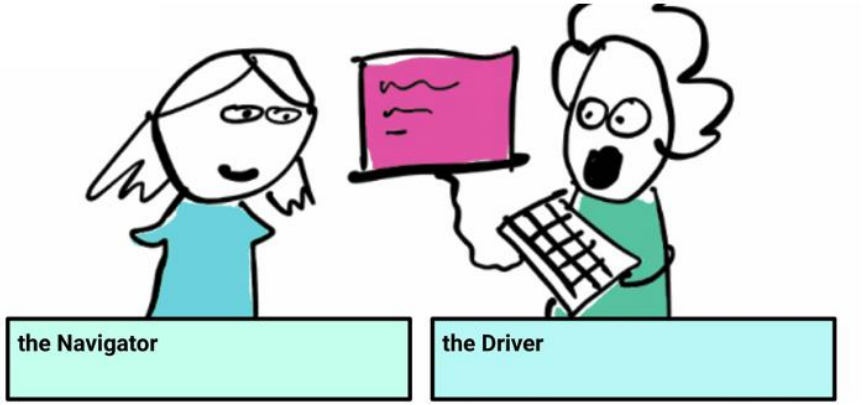
Paired programming



- After 15 mins, swap over!

- We will have a go at this during this week's practical – please arrive on time.

Advantages of paired programming



- Learn useful problem-solving, debugging skills as well as the course material.
- Enables students to work with different partners.
- Learn different approaches to solving the same problems.

Learning objectives

Now, you should be able to:

- Plan a project using pseudocode.
- Explain how variable work within a piece of code.
- Explain the concept of loops.
- Understand and use Booleans.

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Any questions?

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